

FOOD TRENDS UNDERSTANDING CUSTOMER PREFERENCES IN FACEBOOK



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Introduction

- Built an end-to-end analytics solution to understand customer sentiment and product preferences using Facebook feedback.
- Integrated survey, review, and social media data into a unified Power BI model.
- Delivered insights across sentiment, product performance, customer personas, and strategic opportunities.
- Enabled data-driven decision-making for product, marketing, and R&D teams.

Project Objectives

- Consolidate customer feedback from multiple sources into a single analytical environment.
- Identify sentiment trends, recurring complaints, and emerging preferences.
- Segment customers based on behavior, satisfaction, and engagement.
- Provide actionable recommendations for product innovation and marketing strategy.
- Support predictive insights for future campaigns and product launches.

Data Architecture & Tech Stack

- **Data Sources:** Survey forms, social media reviews (Facebook), and product feedback .
- **Processing:** Used Python for text cleaning and VADER for sentiment scoring.
- **Modeling:** Created a semantic model in Power BI using Customer, Product, and Feedback dimensions.

Data Cleaning & Integration

Key Activities

- Imported structured and unstructured data.
- Applied tokenization, normalization, and deduplication.
- Built unified tables with consistent keys (user_id, product_id, sentiment_score).

Outcome

- A clean, standardized dataset ready for advanced analytics.

Sentiment Analysis Dashboard

Key Metrics

- Total Responses: 38K
- Sentiment Distribution: 55% Positive, 32% Negative, 13% Neutral
- Customer Satisfaction Index: 0.55
- Loyal Fan Score: 0.63

Insights

- Strong positive sentiment across major food categories.
- Negative sentiment primarily linked to quality and service issues.
- Keyword analysis highlights recurring themes like “yummy”, “nice”, “good”.

Product Insights Dashboard

Key Metrics

- Avg Quality Rating: 3.93
- Avg Taste Rating: 3.99
- Total Product IDs: 1019

Insights

- Amul leads in brand penetration.
- Premium products show higher retention scores.
- Vegetarian items dominate customer preference.
- Top food items: Chicken Biryani, Burger, Pizza.

Customer Segmentation & Persona

Identified Personas

- Loyal Fans
- Positive Users
- At-Risk Customers
- Silent Users

Behavioral Insights

- Loyal Fans show high engagement across delivery and dining channels.
- At-Risk customers exhibit low sentiment and inconsistent activity.
- Persona distribution varies significantly by age group and region.

Strategy & Innovation Dashboard

Strategic Focus

- Customer-centric marketing aligned with sentiment insights.
- Revenue optimization through high-performing segments.
- Retention planning for at-risk customers.

Innovation Drivers

- Product enhancement using sentiment data.
- Quality improvements based on negative feedback.
- Feature score optimization.

Growth Roadmap

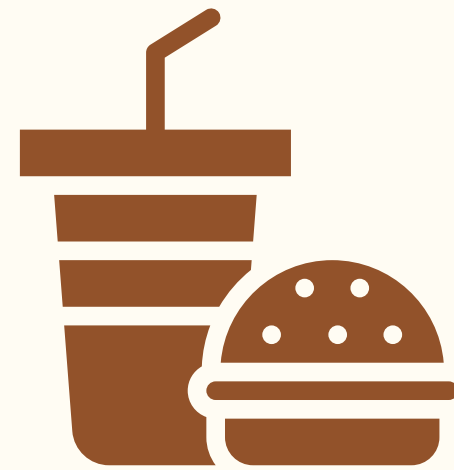
- Expansion into high-sentiment regions.
- Strategic partnerships with digital platforms.
- Loyalty programs to strengthen repeat purchases.
- Predictive analytics for demand forecasting.

Key Outcomes

- Unified sentiment view across platforms and regions.
- Clear visibility into product strengths and weaknesses.
- Customer personas enabling targeted marketing strategies.
- Early detection of emerging trends and issues.
- Actionable recommendations for product, marketing, and R&D teams.

Conclusion

- Facebook feedback is a powerful source of customer intelligence.
- Power BI enables real-time, actionable insights for decision-makers.
- This project strengthens product innovation, customer experience, and strategic planning.



THANK YOU